

HIEU N. NGUYEN

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RESEARCH INTERESTS

I'm interested in (i) understanding the fundamental strengths and limitations of learning algorithms through carefully designed, controlled experiments, (ii) developing intelligent systems with capabilities for reasoning, planning, and creativity, and (iii) leveraging these systems to accelerate scientific discovery and tackle real-world challenges at scale.

Research topics: LLM reasoning, Post-training, AI4Science, Interpretability.

EDUCATION

Pennsylvania State University, University Park

Ph.D in Computer Science

Advisor: Prof. Rui Zhang

Awarded *University Graduate Fellowship*

Aug. 2025 - Current

State College, PA

Vietnam National University, Hanoi

B.S Honors Program in Computer Science

Thesis: An application of Neural Tangent Kernel on DEQ Models

Thesis Advisors: Dr. Hoang Thanh-Tung and Dr. Ta Viet Cuong

Aug. 2022

Hanoi, Vietnam

RESEARCH EXPERIENCE

Ph.D Student, The Pennsylvania State University

Aug. 2025 – Current

- Developed a technique to identify critical weights that influence model reasoning behavior.
- Study the impact of training data characteristics on reasoning coverage.

Research Assistant, VinUniversity – CECS

June 2023 – Aug. 2025

- Developed LLM-SRBench, a benchmark designed to assess LLMs for scientific equation discovery.
- Mitigated over-optimization in LLM alignment algorithms.
- Enhanced the efficiency and security of federated recommendation systems.

AI Resident, FPT Software AI Center

May 2021 – June 2023

- Worked on Out-of-Distribution (OOD) detection, clean-label backdoor attacks, and labeling error detection.

SELECTED PREPRINTS & PUBLICATIONS & WORKSHOPS

*: equal contribution. For my latest research papers, see my Google Scholar profile.

[Agent-Based Auditing of Mathematical Proofs in Research Papers](#)

Hieu N. Nguyen and Rui Zhang

ICML Workshop AI4Research 2026 (Oral presentation)

[Why Do Reasoning Models Lose Coverage? The Role of Data and Forks in the Road](#)

Hieu N. Nguyen*, Parshin Shojaee*, Phuc Minh Nguyen, Nan Zhang, Chandan K Reddy, Khoa D Doan, and Rui Zhang

arXiv 2026

When Reasoning Meets Compression: Understanding the Effects of LLMs Compression on Large Reasoning Models

Nan Zhang, Eugene Kwek, Yusen Zhang, **Hieu N. Nguyen**, Prasenjit Mitra, and Rui Zhang
ICLR 2026

Wicked Oddities: Selectively Poisoning for Effective Clean-Label Backdoor Attacks

Nguyen Hung-Quang, **Hieu N. Nguyen**, The-Anh Ta, Thanh Nguyen-Tang, Kok-Seng Wong, Hoang Thanh-Tung, and Khoa D Doan
ICLR 2025

Mitigating Reward Over-optimization in Direct Alignment Algorithms with Importance Sampling

Nguyen Minh Phuc, **Hieu N. Nguyen**, Duy Minh Ho Nguyen, Anji Liu, An Mai, Binh T. Nguyen, Daniel Sonntag, and Khoa D Doan
NeurIPS 2025

LLM-SRBench: A New Benchmark for Scientific Equation Discovery with Large Language Models

Parshin Shojaei*, **Hieu N. Nguyen***, Kazem Meidani, Amir Barati Farimani, Khoa D Doan, and Chandan K. Reddy
ICML 2025 (Oral presentation - Top 1%)

Towards Efficient Communication and Secure Federated Recommendation System via Low-rank Training

Hieu N. Nguyen, Tuan-Anh Nguyen, Tuan Minh Nguyen, Vu Tien Hoang, Dung D. Le, and Kok-Seng Wong
WWW 2024 (Oral presentation)

Class-based Influence Functions for Error Detection

Nguyen-Duc Thang, Hoang Thanh-Tung, Quan Hung Tran, Dang Huu-Tien, **Hieu N. Nguyen**, Anh T. V. Dau, and Nghi D. Q. Bui
ACL 2023

UNDER REVIEW

On the Primacy Bias in RLVR Training

Nguyen Minh Phuc, **Hieu N. Nguyen**, Parshin Shojaei, Chinh Duc La, Duy Minh Ho Nguyen, Rui Zhang, Khoa D. Doan, Ting Hua, and Nitesh V. Chawla
Submitted to NeurIPS 2026

ACADEMIC SERVICES

Reviewer - SPIGM @ ICML 2026, ICLR 2026, ICCV 2025, NeurIPS 2025, ICLR 2025, IEEE Transactions on Image Processing 2025, NeurIPS 2024, AAAI 2024, ICML 2024.

TEACHING

Intro to AI, Teaching Assistant, Spring 2025

VinUniversity-CECS

Data Mining, Teaching Assistant, Spring 2024, Fall 2024

VinUniversity-CECS

SKILLS

Programming

Python, C/C++, CUDA.

ML frameworks

PyTorch, Jax, DeepSpeed, FSDP, verl, Unsloth.

DevOps

Linux, Shell (Bash/Zsh), Git, Docker, AWS, Claude Code.